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Variable Gain Super-Twisting Sliding Mode Control

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***Abstract*—In this note, a novel, Lyapunov-based, variable-gain super-twisting algorithm (STA) is proposed. It ensures for linear time invariant systems the global, finite-time convergence to the desired sliding surface, when the matched perturbations/uncertainties are Lipschitz-continuous functions of time, that are bounded, together with their derivatives, by known functions. The proposed algorithm has similar properties to the variable-gain first-order sliding mode control, but it provides alleviation to the chattering phenomenon. The results are verified experimentally.**

***Index Terms*—Chattering effect, discontinuous systems, Lyapunov functions, robust control, second-order sliding modes.**

I. Introduction

1) Motivation: Sliding mode (SM) control is a well-known tool for rejecting matched uncertainties/disturbances (see [7], [20], [21], for example). Usually the sliding mode control design (see [20], [21, Ch. 5]) consists of two steps:

- Construction of the desired sliding surface;
- SM enforcement.

In the first generations of SM controllers either relay controllers or unit controllers were used ([9], [20], [21]). The main disadvantage of these control strategies is the so-called "chattering effect" (see [20] and [21], for example).

The three main approaches to chattering alleviation and attenuation in SM systems were proposed in the mid-1980's:

- The use of saturation control instead of the discontinuous one [5], [19]. This approach allows for control continuity but cannot restrict the system dynamics onto the switching surface. It only ensures the convergence to a boundary layer of the sliding manifold, whose size is defined by the slope of the saturation characteristics.
- The observer-based approach [4]. This method allows to bypass the plant dynamics by the chattering loop. This approach reduces the problem of robust control to that of exact and robust estimation.

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- The second-order sliding-mode (SOSM) approach [2], [10], [11], [14]–[16], [18]. It allows for finite-time convergence to zero of not only the sliding variable but its derivative as well.

All SOSM controllers, except for the super-twisting algorithm (STA), require the knowledge of the values of the derivatives (or the special device identifying the maximum value of the output in the case of the suboptimal algorithm [2]), i.e., the knowledge of the perturbation if the sliding surface has relative degree 1 w.r.t. the control ([21]). The STA is a unique *absolutely continuous* sliding-mode algorithm (see, for example, [10]) ensuring all the main properties of first-order sliding mode control for systems with *Lipschitz continuous* matched uncertainties/disturbances with bounded gradients.

Due to the boundedness of the first-order (constant gain) SM control (the sign} function), it is possible to compensate a bounded (matched) perturbation only if the perturbation bound is known *a priori*. The *Variable Gain* first-order SM, for which the gain is adjusted according to an actual bound of the perturbation, has been introduced [20], [21] in order to reduce the magnitude of the chattering effect, and/or to compensate perturbations whose bound can only be calculated on-line or it depends on the state. In this case, a simple Lyapunov function shows that if the gain is selected as a linear function of the perturbation bound finite-time, robust convergence to the sliding manifold will be achieved. It seems also reasonable to introduce variable gains in the STA in order to compensate perturbations, whose gradient bounds are state dependent and/or known on-line, and for which an *a priori* bound is not known. Although it is clear that the gains have to be selected according to the actual bounds of the perturbations it is not obvious how to select the two gains of the STA, so that finite-time and robust convergence to the sliding manifold is achieved. In fact, there are no results in the literature dealing with this problem.

A further drawback of the standard (constant gains) STA is that it does not allow to compensate uncertainties/disturbances growing in time or together with the state variables. This means that the standard STA cannot ensure the sliding motion, even for systems for which the linear part is not exactly known. This restriction of the standard STA is due to the growth behavior of the correction terms, and it is intimately connected with its *homogeneity*, meaning that its defining vector field is homogeneous (see [1], [11], and (13)). Including linear correction terms to the STA would allow the compensation of stronger perturbations far from the sliding surface. However, the obtained algorithm is non homogeneous.

Unfortunately, the proofs of convergence for the STA ([10]), based on the idea of majorant curves, those based on the Lyapunov functions proposed in [17] and [21], or the homogeneity-based proofs [11], can be hardly used for these STA modifications, i.e., adding linear correction terms and/or introducing variable gains.

The goal of this technical note is to propose for the first time a nonhomogeneous, variable gain extension of the standard STA for the standard two-step SM control design procedure using the global variable-gain STA (VGSTA), providing the exact compensation of smooth uncertainties/disturbances bounded, together with their derivatives, by known functions.

2) Contribution: In this technical note we extend the traditional two-step SM design ([20], [21, Ch. 5]) compensating exactly *Lipschitz continuous* uncertainties/disturbances, which are bounded (almost everywhere), together with their derivatives, by known functions. The control algorithm is not homogeneous, has variable gains and it has also additional linearly growing correction terms. The selection of appropriate perturbation-bound dependent gains for the STA is made based on a non classical, time-invariant, non smooth Lyapunov function, that also prove the finite-time convergence. This Lyapunov function is based on [13], where it has been used for the constant gains STA. Furthermore, we also provide:

- the characterization of the class of uncertainties/disturbances that are rejected by the VGSTA, in contrast to the first-order SM, and
- an estimation of the convergence time.

Finally, the theoretically derived results are validated experimentally.

II. Problem Statement

Consider an uncertain linear time invariant system (LTI)

$$\dot{x} = Ax + B(u + f(x,t)) \tag{1}$$

where $x \in \mathbb{R}^n$ is the state vector, $u \in \mathbb{R}^m$ is the control input, A, B are constant matrices of appropriate dimensions, and f is an absolutely continuous uncertainty/disturbance in (1).

Assume the following:

(A1) $\operatorname{rank} B = m$;

(A2) the pair (A, B) is controllable;

(A3) the function f and its gradient are bounded, by known continuous functions, almost everywhere.

It is well known that, under assumptions (A1) and (A2), after the linear state transformation

$$\begin{aligned} \begin{pmatrix} \eta \\ \xi \end{pmatrix} &= Tx, \\ T &= \begin{bmatrix} B^{\perp} \\ B^{+} \end{bmatrix} \\ B^{+} &= (B^T B)^{-1} B^T, \\ B^{\perp} B &= 0 \end{aligned}$$

(1) has the regular form [12]

$$\begin{aligned} \dot{\eta} &= A_{11}\eta + A_{12}\xi \\ \dot{\xi} &= A_{21}\eta + A_{22}\xi + u + \tilde{f}(\eta, \xi, t) \end{aligned} \tag{2}$$

where $\eta \in \mathbb{R}^{n-m}$ and $\xi \in \mathbb{R}^m$. For simplicity of the presentation we will restrict ourselves to the single input case $(m = 1)$, but the results can be easily extended to the multiinput case [20]. For simplicity we will consider only the case when the coefficient of the control variable in (2) is 1, i.e., that the matrix B is completely known. In the case that in (2) the control related term is $b(t)u$, where the coefficient $b(t)$ is uncertain, but positive, i.e., $b(t) \geq b_m$, it suffices to consider in what follows that this coefficient is constant, and equal to b_m. Since it is always possible to normalize the equation, so that the coefficient of u is the unity, we are brought back to (2).

The main idea of sliding mode control is to design the sliding variable

$$s = \xi - K\eta \tag{3}$$

such that, when the motion is restricted to the manifold $s = 0$, the reduced-order model

$$\dot{\eta} = (A_{11} + A_{12}K)\eta \tag{4}$$

has the required performance. Since the pair (A_{11}, A_{12}) is controllable, matrix K can be designed using any linear control design method for system (4), as, for example: (i) eigenvalue assignment, or (ii) optimal control (LQR), a strategy known as the optimal sliding manifold design.

Using (η^T, s) as state variables, and applying the controller

$$\begin{aligned} u = -(A_{21} + A_{22}K - K(A_{11} + A_{12}K))\eta - \\ -(A_{22} - KA_{12})s + v \end{aligned} \tag{5}$$

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(2) takes the form

$$\begin{aligned}\dot{\eta} &= (A_{11} + A_{12}K)\eta + A_{12}s,\\ \dot{s} &= v + \tilde{f}(\eta, s + K\eta, t).\end{aligned} \tag{6}$$

When the perturbation is bounded by a known function $\varrho(x)$

$$|f(x,t)| \leq \varrho(x), \tag{7}$$

the (first order) sliding mode can be enforced by a variable-gain controller

$$v = -(\varrho(x) + \varrho_0)\text{sign}(s) \tag{8}$$

with $\varrho_0 > 0$. The main disadvantage of this controller is that it produces the *chattering effect*, which grows along with the uncertainty bound $\varrho(x)$.

In Section III, we extend the Lyapunov based design [13] of the STA in order to include: (i) linear terms, and (ii) variable gains to alleviate the drawbacks of the standard STA. The use of the Lyapunov method is instrumental here, since neither the geometric nor the homogeneity proofs of the standard STA allow to deal with these extensions.

III. The VGSTA

The VGSTA proposed here is given by [13]

$$v = -k_1(t,x)\phi_1(s) - \int_0^t k_2(t,x)\phi_2(s)dt \tag{9}$$

where

$$\begin{aligned}\phi_1(s) &= |s|^{1/2}\text{sign}(s) + k_3 s,\ k_3 > 0\\ \phi_2(s) &= \phi_1'(s)\phi_1(s)\\ &= \frac{1}{2}\text{sign}(s) + \frac{3}{2}k_3|s|^{1/2}\text{sign}(s) + k_3^2 s.\end{aligned}$$

When $k_3 = 0$ and the gains k_1 and k_2 are constant, we recover the standard STA. $k_3 > 0$ allows to deal with perturbations growing linearly in s, i.e., outside of the sliding surface, and the variable gains k_1 and k_2 make it possible to render the sliding surface insensitive to perturbations growing with bounds given by known functions. Note that the control variable is absolutely continuous, in contrast to the discontinuous character of the SMC (8).

To provide a description of the perturbations supported by the VGSTA, note that the uncertainty/disturbance can always be written as

$$\begin{aligned}\tilde{f}(\eta, s + K\eta, t) &= \underbrace{[\tilde{f}(\eta, s + K\eta, t) - \tilde{f}(\eta, K\eta, t)]}_{g_1(\eta,s,t)}\\ &\quad + \underbrace{\tilde{f}(\eta, K\eta, t)}_{g_2(\eta,t)}\\ \tilde{f}(\eta, s + K\eta, t) &= g_1(\eta, s, t) + g_2(\eta, t)\end{aligned} \tag{10}$$

where $g_1(\eta, s, t) = 0$, when $s = 0$. Due to its continuity the VGSTA is not able to compensate all the perturbations in the class given by (7), i.e., when g_1 and g_2 are just measurable and bounded functions of time, but in accordance with assumption (A3), it is insensitive to uncertainties/disturbances $f(x,t)$ satisfying (almost everywhere)

$$\begin{aligned}|g_1(\eta,s,t)| &\leq \varrho_1(t,x)|\phi_1(s)|\\ &= \varrho_1(t,x)[1 + k_3|s|^{1/2}]|s|^{1/2}\\ \left|\frac{d}{dt}g_2(\eta,t)\right| &\leq \varrho_2(t,x)|\phi_2(s)|\\ &= \frac{1}{2}\varrho_2(t,x) +\\ &\quad + k_3\varrho_2(t,x)\left[\frac{3}{2} + k_3|s|^{1/2}\right]|s|^{1/2}\end{aligned} \tag{11}$$

where $\varrho_1(t,x) \geq 0$, $\varrho_2(t,x) \geq 0$ are known continuous functions. Note that, although the component g_1 of the perturbation has to vanish when $s = 0$, this is not the case for the derivative of the component g_2 of the perturbation, since for $s = 0$ it is sufficient that this derivative is bounded by $(1/2)\varrho_2(t,x)$ [see (11)].

System (6), driven by the VGSTA (9), can be written as

$$\begin{aligned}\dot{\eta} &= (A_{11} + A_{12}K)\eta + A_{12}s,\\ \dot{s} &= -k_1(t,x)\phi_1(s) + z + g_1(\eta,s,t)\\ \dot{z} &= -k_2(t,x)\phi_2(s) + \frac{d}{dt}g_2(\eta,t).\end{aligned} \tag{12}$$

The solutions of the discontinuous differential equations and inclusions are understood in the sense of Filippov [8]. Recall [1], [11] that a vector field $f : \mathbb{R}^n \to \mathbb{R}^n$ is called *homogeneous of the degree* $q \in \mathbb{R}$ *with the dilatation* $d_\kappa : (x_1, x_2, \ldots, x_n) \mapsto (\kappa^{m_1}x_1, \kappa^{m_2}x_2, \ldots, \kappa^{m_n}x_n)$, where $m_1, m_2, \ldots, m_n$ are some positive numbers, if for any $\kappa > 0$ the identity holds

$$f(x) = \kappa^{-q}d_\kappa^{-1}f(d_\kappa x). \tag{13}$$

When $k_3 = 0$ and the gains k_1 and k_2 are constant, the STA is *homogeneous* since for the vector field of subsystem (s,z) in (12), with $g_1 = g_2 = 0$, (13) holds with $m_1 = 1, m_2 = 1/2, q = -1/2$. For the VGSTA however, when $k_3 > 0$, or the gains k_1, k_2 depend on (s,z), the homogeneity condition is not fulfilled.

The main result of the technical note is as follows.

Theorem 1: Suppose that for some known continuous functions $\varrho_1(t,x) \geq 0$, $\varrho_2(t,x) \geq 0$ the inequalities (11) are satisfied. Then for any initial condition (η_0^T, s_0, z_0) the sliding surface $s = 0$ will be reached in finite time if the variable gains are selected as

$$\begin{aligned}k_1(t,x) &= \delta + \frac{1}{\beta}\left\{\frac{1}{4\epsilon}[2\epsilon\varrho_1 + \varrho_2]^2 a + 2\epsilon\varrho_2 + \epsilon + \right.\\ &\quad \left. + [2\epsilon + \varrho_1](\beta + 4\epsilon^2)\right\}\\ k_2(t,x) &= \beta + 4\epsilon^2 + 2\epsilon k_1(t,x)\end{aligned} \tag{14}$$

where $\beta > 0$, $\epsilon > 0$, $\delta > 0$ are arbitrary positive constants. The reaching time of the sliding surface can be estimated by

$$T = \frac{2}{\gamma_2}\ln\left(\frac{\gamma_2}{\gamma_1}V^{1/2}(s_0,z_0) + 1\right) \tag{15}$$

where $V(s,z) = \zeta^T P\zeta$, $\zeta^T = [\,|s|^{1/2}\text{sign}(s) + k_3 s, \quad z\,]$, and γ_1 and γ_2 can be found from (19). ■

In Section IV the proof of this result will be given using a Lyapunov approach.

Theorem 1 proposes a methodology to design a sliding mode controller ensuring a sliding motion on the surface $s = 0$ (3), substituting the *discontinuous* control variable from the SMC (8) by an *absolutely continuous* control provided by the VGSTA (9). The continuity of the control of the VGSTA has as a consequence that: i) the chattering level can be substantially reduced [3]; and ii) the class of perturbations supported by the VGSTA, given by (11), is different from the class of perturbations (7) to which the SMC is insensitive. For example, when the bounding functions $\rho(x)$ in (7), and $\rho_1(x)$, $\rho_2(x)$ in (11) are constant, the SMC (8) is insensitive to possibly nondifferentiable but bounded perturbations. The STA however, can support unbounded perturbations but with a bounded (time) derivative. In the variable-gain case considered here these bounds can be given by known functions of time and/or the states. It is important to notice that, since the gains in the algorithm

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(12) are time-varying and state dependent, the algorithm is non-homogeneous, and so the classical homogeneity arguments cannot be used to ascertain the stability. Moreover, the stability of the STA for every frozen value of time and state is clearly not sufficient for the stability of (12). The existence of a Lyapunov function for this system becomes crucial, a fact shown for the first time in the following proof of the theorem. Moreover, the fact that the Lyapunov function proposed for the proof is time-invariant is far from obvious.

PROOF OF THEOREM 1

The proof of Theorem 1 will be done using a strict Lyapunov function for the subsystem (s,z) of (12). Extending the ideas of [13] we will show that the quadratic form

$$V(s,z)=\zeta^T P\zeta \tag{16}$$

where

$$\zeta^T=[\,|s|^{1/2}\mathrm{sign}(s)+k_3 s,\quad z\,] \tag{17}$$

$$P=\begin{bmatrix}p_1 & p_3\\ p_3 & p_2\end{bmatrix}=\begin{bmatrix}\beta+4\epsilon^2, & -2\epsilon\\ -2\epsilon & 1\end{bmatrix}, \tag{18}$$

with arbitrary positive constants $\beta>0$, $\epsilon>0$, is a *time-invariant, global, robust and strong (strict) Lyapunov function* for the subsystem (s,z) of (12), showing finite time convergence. It is *time-invariant*, since matrix P in (16) is constant. It is *robust*, since its derivative is negative definite for all the allowed perturbations. Function (16) is positive definite, everywhere continuous and differentiable everywhere except on the set $\mathcal{S}=\{(s,z)\in R^2 \mid s=0\}$.

The inequalities (11) can be rewritten as $g_1(\eta,s,t)=\alpha_1(t,x)\phi_1(s)$ and $(d/dt)g_2(\eta,t)=\alpha_2(t,x)\phi_2(s)$ for some functions $|\alpha_1(t,x)|\le \varrho_1(t,x)$ and $|\alpha_2(t,x)|\le \varrho_2(t,x)$. Using these functions and noting that $\phi_2(s)=\phi_1'(s)\phi_1(s)$ one can show that

$$\begin{aligned}\dot\zeta&=\begin{bmatrix}\phi_1'(s)\{-k_1(t,x)\phi_1(s)+z+g_1(x,t)\}\\ -k_2(t,x)\phi_2(s)+\frac{d}{dt}g_2(x,t)\end{bmatrix}\\ &=\phi_1'(s)\begin{bmatrix}-(k_1(t,x)-\alpha_1(t,x)), & 1\\ -(k_2(t,x)-\alpha_2(t,x)) & 0\end{bmatrix}\zeta\\ &=\phi_1'(s)\mathcal{A}(t,x)\zeta\end{aligned}$$

for every point in $\mathbb{R}^2\backslash\mathcal{S}$, where this derivative exists. Similarly one can calculate the derivative of $V(x)$ on the same set as

$$\begin{aligned}\dot V(s,z)&=\phi_1'(s)\zeta^T(\mathcal{A}^T(t,x)P+P\mathcal{A}(t,x))\zeta\\ &=-\phi_1'(s)\zeta^T Q(t,x)\zeta\end{aligned}$$

where

$$Q=\begin{bmatrix}2(k_1(t,x)-\alpha_1)p_1+2(k_2(t,x)-\alpha_2)p_3 & \bigstar\\ (k_1(t,x)-\alpha_1)p_3+(k_2(t,x)-\alpha_2)p_2-p_1, & -2p_3\end{bmatrix}$$

where $\bigstar$ is used to indicate a symmetric element. Selecting P as in (18) and the gains as in (14), we have (the arguments of the functions were left out)

$$\begin{aligned}&Q-2\epsilon\mathbb{I}\\ &=\begin{bmatrix}2\beta k_1+4\epsilon(2\epsilon k_1-k_2)-2(\beta+4\epsilon^2)\alpha_1+4\epsilon\alpha_2-2\epsilon & \bigstar\\ k_2-2\epsilon k_1-(\beta+4\epsilon^2)+2\epsilon\alpha_1-\alpha_2, & 2\epsilon\end{bmatrix}\\ &=\begin{bmatrix}2\beta k_1-(\beta+4\epsilon^2)(4\epsilon+2\alpha_1)+4\epsilon\alpha_2-2\epsilon & \bigstar\\ 2\epsilon\alpha_1-\alpha_2, & 2\epsilon\end{bmatrix}\end{aligned}$$

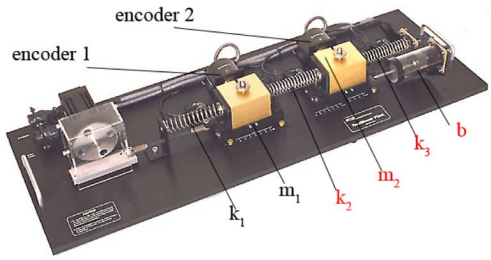


Fig. 1. The experimental system.

that is positive definite for every value of (t,x). This shows that

$$\begin{aligned}\dot V&=-\phi_1'(s)\zeta^T Q(t,x)\zeta\le -2\epsilon\phi_1'(s)\zeta^T\zeta\\ &=-2\epsilon\left(\frac{1}{2|s|^{1/2}}+k_3\right)\zeta^T\zeta.\end{aligned}$$

Since $\lambda_{\min}\{P\}\|\zeta\|_2^2\le\zeta^T P\zeta\le\lambda_{\max}\{P\}\|\zeta\|_2^2$, where $\|\zeta\|_2^2=\zeta_1^2+\zeta_2^2=|s|+2k_3|s|^{3/2}+k_3^2s^2+z^2$ is the Euclidean norm of ζ, and

$$|\zeta_1|\le\|\zeta\|_2\le\frac{V^{1/2}(s,z)}{\lambda_{\min}^{1/2}\{P\}}$$

we can conclude that

$$\begin{aligned}\dot V&\le-\gamma_1 V^{1/2}(s,z)-\gamma_2 V(s,z)\\ \gamma_1&=\frac{\epsilon\lambda_{\min}^{1/2}\{P\}}{\lambda_{\max}\{P\}},\ \gamma_2=\frac{2\epsilon k_3}{\lambda_{\max}\{P\}}.\end{aligned} \tag{19}$$

Note that the trajectories of the STA cannot stay on the set $\mathcal{S}=\{(s,z)\in R^2\mid s=0\}$. This means that V is a continuously decreasing function and using a Lyapunov's Theorem for Differential Inclusions [6, Proposition 14.1, p. 205] (that does not require differentiability of the Lyapunov function) we can conclude that the equilibrium point $(s,z)=0$ is reached in finite time from every initial condition. We note that classical Lyapunov theorems cannot be used here, since they require: (i) a continuous right hand side of the differential equation and/or (ii) a differentiable (or at least locally Lipschitz continuous) Lyapunov function candidate. None of these conditions is satisfied here. The result in [6, Proposition 14.1, p. 205] is valid for Differential Inclusions and it only requires continuity of the Lyapunov function candidate.

Since the solution of the differential equation

$$\dot\nu=-\gamma_1\nu^{1/2}-\gamma_2\nu,\quad \nu(0)=\nu_0\ge 0$$

is given by

$$\nu(t)=\exp(-\gamma_2 t)\left[\nu_0^{1/2}+\frac{\gamma_1}{\gamma_2}\left(1-\exp\left(\frac{\gamma_2}{2}t\right)\right)\right]^2.$$

It follows that $(s(t),z(t))$ converges to zero in finite time and reaches that value at most after a time given by (15). This concludes the proof of Theorem 1.

IV. EXPERIMENTAL STUDY: THE MASS-SPRING-DAMPER SYSTEM

A. Model Description

A mass-spring-damper (MSD) system, model 210a from Educational Control Products (ECP) (see Fig. 1), consisting of 2 masses, 3 springs, 1 damper, and a DC motor in the configuration shown in Fig. 2(a), has been used.

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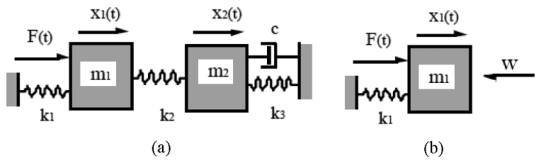


Fig. 2. Left: the MSD system. Right: the mass-spring system with disturbance.

The dynamics of the system is given by

$$\begin{aligned}\dot{x}_1 &= x_2,\\ \dot{x}_2 &= -\frac{\kappa_1}{m_1}x_1 - \frac{\kappa_2}{m_1}x_1 + \frac{\kappa_2}{m_1}x_3 + \frac{1}{m_1}u\\ \dot{x}_3 &= x_4\\ \dot{x}_4 &= -\frac{(\kappa_3+\kappa_2)}{m_2}x_3 - \frac{c_1}{m_2}x_4 + \frac{\kappa_2}{m_2}x_1\end{aligned} \tag{20}$$

where x_1, x_2 (x_3, x_4) are the position and velocity of mass 1 (mass 2), and the control $u = F$ is the force that the DC motor applies to mass 1. Both positions x_1, x_3 can be measured using encoders. Nominal values for the parameters are

$$\begin{aligned}m_1 &= 1.28\ [\mathrm{kg}]\\ m_2 &= 1.05\ [\mathrm{kg}],\\ \kappa_1 &= 190\ \left[\frac{\mathrm{N}}{\mathrm{m}}\right]\\ \kappa_2 &= 780\ \left[\frac{\mathrm{N}}{\mathrm{m}}\right]\\ \kappa_3 &= 450\ \left[\frac{\mathrm{N}}{\mathrm{m}}\right]\\ c_1 &= 15\ \left[\frac{\mathrm{N}\cdot\mathrm{s}}{\mathrm{m}}\right].\end{aligned}$$

The control objective is to maintain the position of mass 1 fixed at $x_1 = x_d$, despite of the behavior of mass 2, that can be considered as a perturbation for the controller design [see Fig. 2(b)]. The change of coordinates

$$\eta = \frac{1}{m_1}(x_1 - x_d),\ \xi = m_1 x_2$$

brings the mass1 system to the regular form

$$\dot{\eta} = \frac{1}{m_1^2}\xi, \quad \dot{\xi} = -\kappa_1 m_1 \eta - \kappa_1 x_d + u + w \tag{21}$$

where $w = \kappa_2(x_3 - x_1)$ is the "perturbing" effect of mass 2 on mass 1.

B. Control Design

The linear sliding surface

$$s = \xi + K\eta \tag{22}$$

leads to a linear reduced-order dynamics $\dot{\eta} = -K/m_1^2\eta$, so that the origin of (21) will be reached exponentially fast, and with a desired performance. Using as state variables (η, s), and applying the equivalent control $u = -K/m_1^2(-K\eta + s) + \kappa_1 m_1 \eta + \kappa_1 x_d + v$, (21) becomes

$$\begin{aligned}\dot{\eta} &= \frac{1}{m_1^2}(-K\eta + s),\\ \dot{s} &= v + w\end{aligned} \tag{23}$$

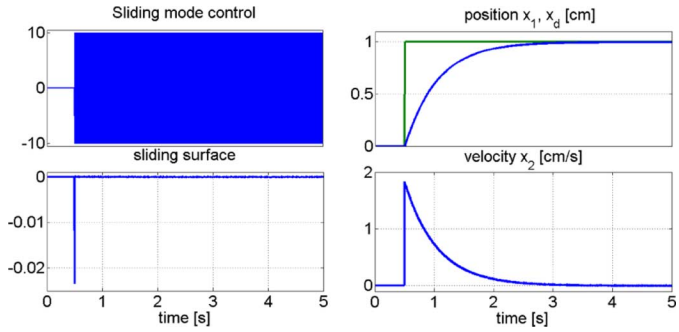


Fig. 3. Simulation results for the constant gain first-order sliding mode control.

where the disturbance term is

$$w(t,\eta) = \kappa_2 x_3 - \frac{\kappa_2}{m_1}\eta. \tag{24}$$

In the decomposition (10) terms g_1 and g_2 are given by

$$\begin{aligned}g_1(\eta, s, t) &= 0,\\ g_2(\eta, t) &= \kappa_2 x_3(t) - \frac{\kappa_2}{m_1}\eta,\\ \frac{dg_2(\eta,t)}{dt} &= \kappa_2 x_4 - \frac{\kappa_2}{m_1^3}(s - K\eta)\end{aligned} \tag{25}$$

and therefore the bounds on the perturbations (11) are fulfilled with

$$\begin{aligned}\varrho_1(t,x) &= 0,\\ \varrho_2(t,x) &= \max\left\{2\kappa_2\left(|x_4(t)| + \frac{K}{m_1^3}|\eta|\right), \frac{\kappa_2}{k_3^2 m_1^3}\right\}.\end{aligned} \tag{26}$$

C. Simulation Results

For comparison a sliding surface (22), with $K = 3$, was designed, and three different controllers were used for the control variable v in (23) and tested in simulation:

- A first-order SM control with fixed gain $v = -L\mathrm{sign}(s)$, where L has been selected as $L = 10$.
- A standard STA with constant gains. Its structure is given by (9), with $k_3 = 0$, and constant gains given by

$$k_1 = 1.5\sqrt{\dot{w}_{\max}},\ k_2 = 1.1\,\dot{w}_{\max}$$

  and $\dot{w}_{\max}$ is an upper bound of the derivative of the perturbation given by $\dot{w}_{\max} = \kappa_2(|x_{4\max}| + |x_{2\max}|)$. Via simulation with the SM controller the values $|x_{4\max}| = |x_{2\max}| = 0.02\ [\mathrm{m/s}]$ were obtained.
- The VGSTA given in (9), having linear correction terms, with constant $k_3 = 8$, and variable gains k_1 and k_2 defined by (14), with parameters $\delta = 0.001$, $\beta = 7$, $\epsilon = 0.11$, and ϱ_1, ϱ_2 as given in (26).

For the simulations at time $t = 0.5$ [s] a reference position $x_d = 1\ [\mathrm{cm}]$ was demanded. Fig. 3 and Fig. 4(a) and (b) show the simulation results for the SM controller, the standard STA and the VGSTA, respectively. For the simulations the three algorithms are able to achieve the objective, but as expected the SM controller has a rather strong chattering effect, which is substantially reduced by both STA and VGSTA.

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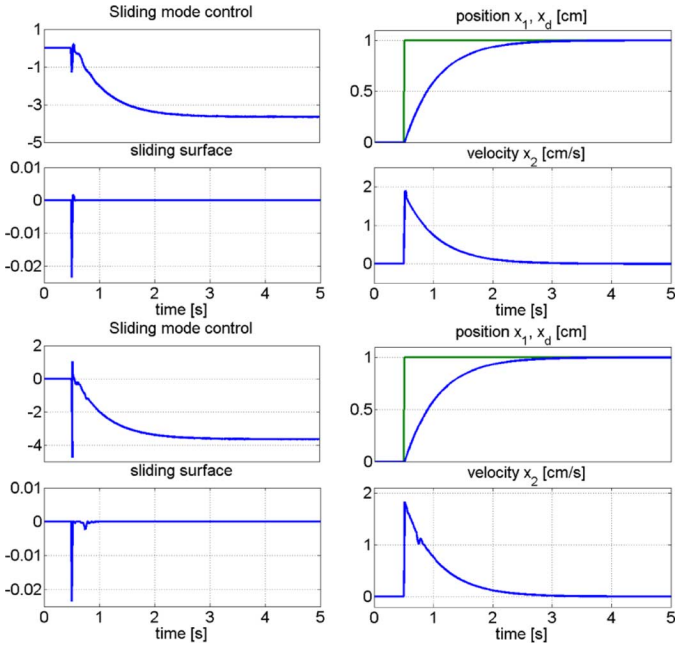


Fig. 4. Simulation results for the Standard, Constant Gains STA (top), and the Variable Gains STA (bottom).

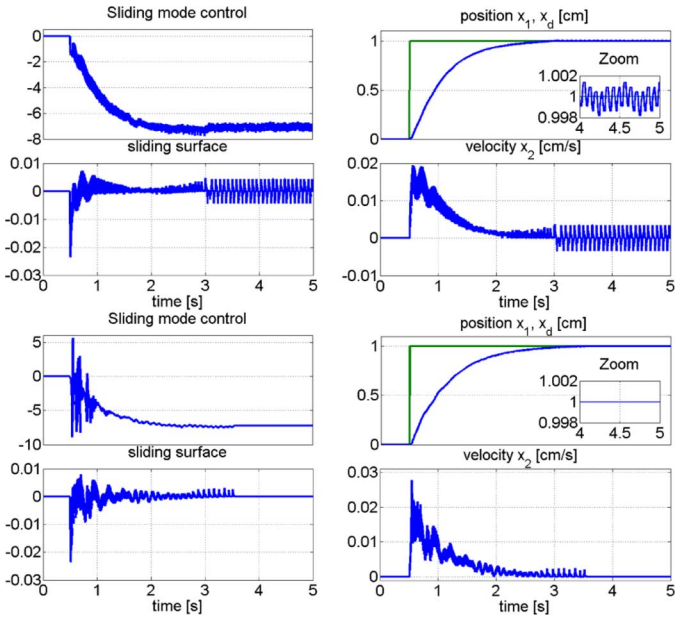


Fig. 5. Experimental results for the Standard, Constant Gains STA (top), and the Variable Gains STA (bottom).

D. Experimental Results

The three controllers (SM, STA, VGSTA), with the same settings as in the simulation study, were tested experimentally. Due to the strong chattering effect of the SM controller, the current protection of the equipment was always activated. The experimental results for the standard STA are shown in Fig. 5(a) and (b) for the VGSTA. Both controllers are able to regulate the position at $x_1 = x_d = 1$ [cm], but the Fixed Gains STA shows a chattering effect after reaching the objective (see zoom in Fig. 5(a)). This chattering effect is further reduced in the VGSTA, as illustrated in the zoom in Fig. 5(b). A video with the experimental results can be consulted at the webpage http://verona.fi-p.unam.mx/~lfridman/mds.php

V. Conclusion

In this note, the classical STA has been modified by introducing extra linear correction terms and time and state varying gains, and it has been termed VGSTA. The VGSTA allows to compensate a larger class of perturbations than the STA and to further reduce the chattering effect of the classical (discontinuous) first-order sliding modes control. Moreover, when online bounds of the perturbations are known, the gains of the VGSTA can be varied accordingly, improving the performance of the algorithm. A Lyapunov function approach has been used to asset the stability and robustness properties of the VGSTA, to estimate its convergence time, and to specify how the variable gains have to be designed to assure this properties.

The proposed VGSTA can substitute the discontinuous sliding mode control by an absolutely continuous control ensuring chattering reduction and the exact compensation of Lipschitz continuous uncertainties/disturbances bounded together with their gradients by known functions almost everywhere. The effectiveness of the proposed design procedures was validated through experimental results using the MSD System.

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